Furnace Control Example

INT0 – PD2 (HOT)
INT1 – PD3 (COLD)
Both need to be inputs

PA.7 – output to solenoid (turn furnace ON/OFF)
Based on temperature waveforms – HOT' sensor shows temperature goes above 20+1 degrees Celsius on falling edge
Based on temperature waveforms – COLD' sensor shows temperature goes below 20-1 degrees Celsius on falling edge
Turn furnace ON when temp is COLD' based on falling edge and OFF when temp is HOT' based on falling edge

```
#include <avr/io.h>
#include <avr/interrupt.h>

ISR(INT0_vect);
ISR(INT1_vect);
void main()
{
	DDRD &= 0b11110011; // PD2, PD3 inputs
	PORTD |= 0b00001100; // pullups for PD2, PD3
	DDRA |= 0b10000000; // PA.7 output to furnace

	PORTA &= 0b00000000; // initialize furnace as off
	// check to see if too cold => if too cold and furnace off, then then cold sensor will never change
	// falling edge trigger for INT0
	// cold sensor used to indicate that furnace should be on
	if (!(PIND & 1<<3))
		PORTA |= 0b10000000; // if cold, then furnace needs to be on

	MCUCR |= 0b00001010; // INT1, INT0 both falling edge
	GICR |= 0b11000000; // enable INT1, INT0 as interrupt sources
	sei();  // global interrupt enable

	while(1); // temp sensors will drive furnace ON/OFF
}

ISR(INT1_vect) {
	PORTA |= 0b10000000; // turn ON furnace
}
ISR(INT0_vect) {
	PORTA &= 0b01111111; // turn OFF furnace
}
```

Intrusion Warning System

INT0 (PD2) – door sensor
Falling edge with door sensor triggers event

400 Hz for 1s on PA0

Assume fosc = 8 MHz

Output waveform f = 400 Hz => T = 0.0025 s or T = 2.5 ms or 2500 us

Thalf = 1250 us
1250 us = n tMC = n (1 s/8 10^6 cyc) (1 cyc/1 MC) => n = 10000 MCs
10000 MCs > 255 so need prescale
10000/64 = 156.25 < 256 so n = 156
n = 156
Use Timer 0 with 64 prescaler

1 s total duration = n tMC
1s = n (1 cc/MC)(1s/8 10^6 cc) => n = 8 10^6 MCs

8,000,000/10,000 = 800 half waves of 400 Hz waveform loop of 800 half waves
OR
Separate timer for 1s duration (Timer 1 – 16 bit)
8,000,000/256 = 31,250 < 65,535 (16 bit) so use 256 prescaler for Timer 1 with TCNT1 = -31,250

Need separate interrupts for Timer 0 and Timer 1

```
#include <avr/io.h>
#include <avr/interrupt.h>
unsigned char done;
void main() {

        DDRA != 0b00000001; // PA0 output
        DDRD &= 0b11111011; // PD2 input
        PORTD |= 0b00000100; // PD2 pullup

        MCUCR = 0b00000010; // INT0 falling edge
        GICR = 1<<INT0; // enable INT0
        SEI(); // global interrupt enable

        While(1);  // Let interrupts take care of intrusion detection and alarm
}


// Timer interrupts are enabled in INT0 with door sensor event
// Alarm 400 Hz waveform for 1s generated with door sensor event
```

```
ISR(INT0_vect) {
	TCNT0 = -156;  // preload for 400 Hz half wave
	TCCR0 = 0b00000011; // TIMER 0 on with 64 prescale

	TCNT1 = -31,250;  // preload for Timer 1
	TCCR1A = 0x00;
	TCCR1B = 0b00000100; // Timer 1 with 256 prescale

	TIMSK = (1<<TOIE1) | (1<<TOIE0);  // enable Timer 1 and 0 interrupt (TOVx) sources
                                          // remain enabled until timer 1 overflow interrupt is executed
}

ISR(Timer1_OVR_vect)
{
// disable timer interrupts and turn off timers
	TCCR0 = 0x00;
	TCCR1B = 0x00; // turn off timers 0 and 1
	TIMSK = 0x00; // turn off Timer 0 and 1 interrupt sources
}

ISR(Timer0_OVR_vect)
{
	PORTA ^= 0b00000001; // toggle PA.0 for 400 Hz waveform
	TCNT0 = -156;
}
```